

PORTABLE BATTERY KIT (PBK)

The PBK is built around a commercially available plastic box from Plano Corporation model #1312-52, a 12v 12Ah LiFePO4 battery with BMS and a management printed circuit board developed by ISV. LED bulb and socket sets are supplied by Global Himalayan Expedition (GHE).



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Each PBK will charge in about 4 hours inside the Sunblazer IV Station box connected to a dedicated charger.



Also inside the Station box is a single Kiosk controller that automatically connects wirelessly to all PBK's when they are plugged into a charger. The Kiosk records amp hours required by each PBK to complete it's charge and can provide billing information to the operator as well as maintaining records of usage.

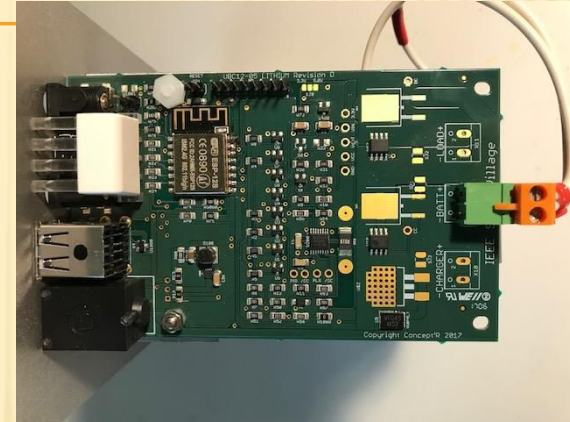


The Kiosk controller is a commercially available component that only requires programming and packaging.

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12 volt 12 Ah LiFePO4 battery
4.125"W x 4.125"H x 2.75"D



Control Printed Circuit Board
mounts to underside of
faceplate



Example of battery mounting
pad.
 $\frac{3}{4}$ " PVC board



Battery pad mounts to bottom of
box

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User interface and graphics

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Portable Battery Kit (PBK) example Bill Of Materials

Part Description/Part No.	Manufacturer/Vendor	QTY	Unit Price	Amount	Notes
Plastic storage box/ # 1312, 11" x 5" x 7"	Plano / Amazon	1	\$4.98	\$4.98	Shipping included
LiFePO4 12v 12 Ah Battery	Shen Zhen Mottcell Battery Technology Co,LtD	1	\$42.50	\$42.50	Not including shipping.
Aluminum faceplate, machined	TBD	1	\$12.00	\$12.00	
Front panel graphics, 1/32" thk aluminum overlay	Printsignsquick.com	1	\$7.16	\$7.16	Shipping included, Options for screen printing faceplate directly
PBK control PCB	TBD	1	\$35.00	\$35.00	Cost based on 24 pc. build
ATO/ATC inline fuse holder, 15A/ #070-675	Parts-Express.com	1	\$1.95	\$1.95	
15 amp ATC Fuse (Pack of 25)/ #ATC15-25	Metra/ Newegg.com	1	\$0.19	\$0.19	Sold in 25 pc. pack for \$4.73 not inclu. shipping
wire splices for fuse and battery, 14-16 AWG/ #31760	Molex/ Waytekwire.com	2	\$0.08	\$0.16	
3" 18 AWG stranded wire, battery connection/ #180C	JT&T/ Wiringdepot.com (QTY in inches)	3	\$0.01	\$0.02	100 ft. roll is \$9.87, 3" required as indicated by QTY column This bracket holds the PCB to front panel
1/16" Thk x 5/8" x 1-1/2" Aluminum angle 3-1/4" long	TBD	1	\$0.50	\$0.50	
4-1/2" x 1/16" Thk x 1-1/2" aluminum bar/ #8975K199	McMaster Carr (QTY 6 ft., priced per inch)	4.5	\$0.11	\$0.50	6 ft. length, priced per inch for total of 4.5 inches required
5"x3.5"x 3/4" PVC Trim board (battery pad)	Lowes Hardware	1	\$0.82	\$0.82	\$23 -12 ft. length yeilds 28 pieces
5"x3.5"x 1/16" steel plate (Battery protection)	M-D Building supply	1	\$1.42	\$1.42	\$17 - 12" x 24" 16 GA yeilds 12 pieces
#12 x 3/4" pan head wood screws	Lowes Hardware	12	\$0.20	\$2.40	
4ft nylon lacing cord (Battery mounting)	Alpha Wire/Newark Electronics	1	\$0.20	\$0.20	\$75 - sold in 1500 ft. roll yeilds 375 4 ft. pieces
Misc. hardware	McMaster Carr	1	\$1.00	\$1.00	PCB standoffs/ nuts
Light Kit					
Two 4W LED bulbs					
Two lamp socket bases					
5 meters low voltage wire w/switches	Global Himalayan Expedition	1	\$7.00	\$7.00	Not inclu. shipping
DC Power Connector 2.5mm/ #490-PP-002B	Mouser Electronics	1	\$0.91	\$0.91	Not inclu. shipping
Labor est. 2 hours with machining			0	0	
Grand Total				\$118.71	Without labor